COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 08

Sequential Circuit



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Recap of Previous Lecture



- Synch. Counter
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Synch. Counter design Cont.

Design a Counter with sequence → (0-1-4-2-5) using TFF.

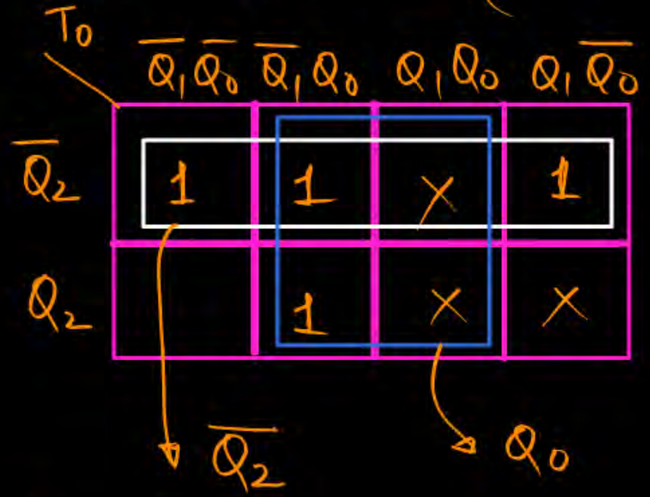
→ MOD no. = 5 → FF → 3

	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
	0	0	0	0	0	1	0	0	1
	0	0	1	1	0	0	1	0	1
	0	1	0	1	0	1	1	1	1
(3)	0	1	1	X	X	X	X	X	X
	1	0	0	0	1	0	1	1	0
	1	0	1	0	0	0	1	0	1
(6)	1	1	0	X	X	X	X	X	X
(7)	1	1	1	X	X	X	X	X	X

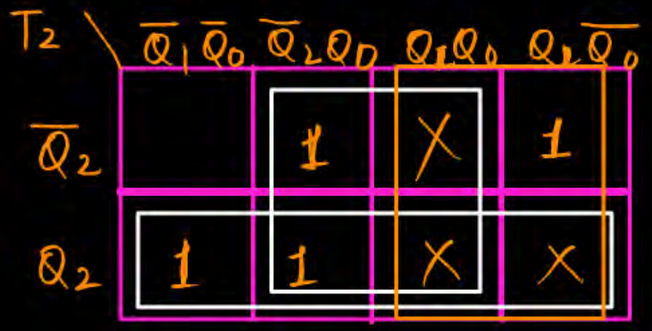
$$T_2[Q_2, Q_1, Q_0] = \Sigma(1,2,4,5) + d\Sigma(3,6,7)$$

$$T_1[Q_2, Q_1, Q_0] = \Sigma(2,4) + d\Sigma(3,6,7)$$

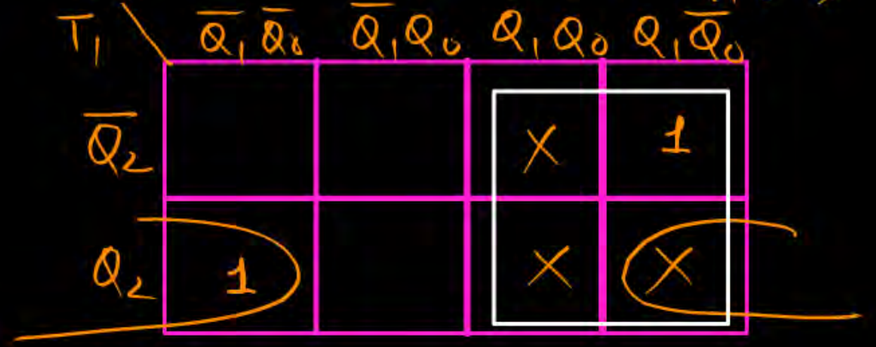
$$T_0[Q_2, Q_1, Q_0] = \Sigma(0,1,2,5) + d\Sigma(3,6,7)$$



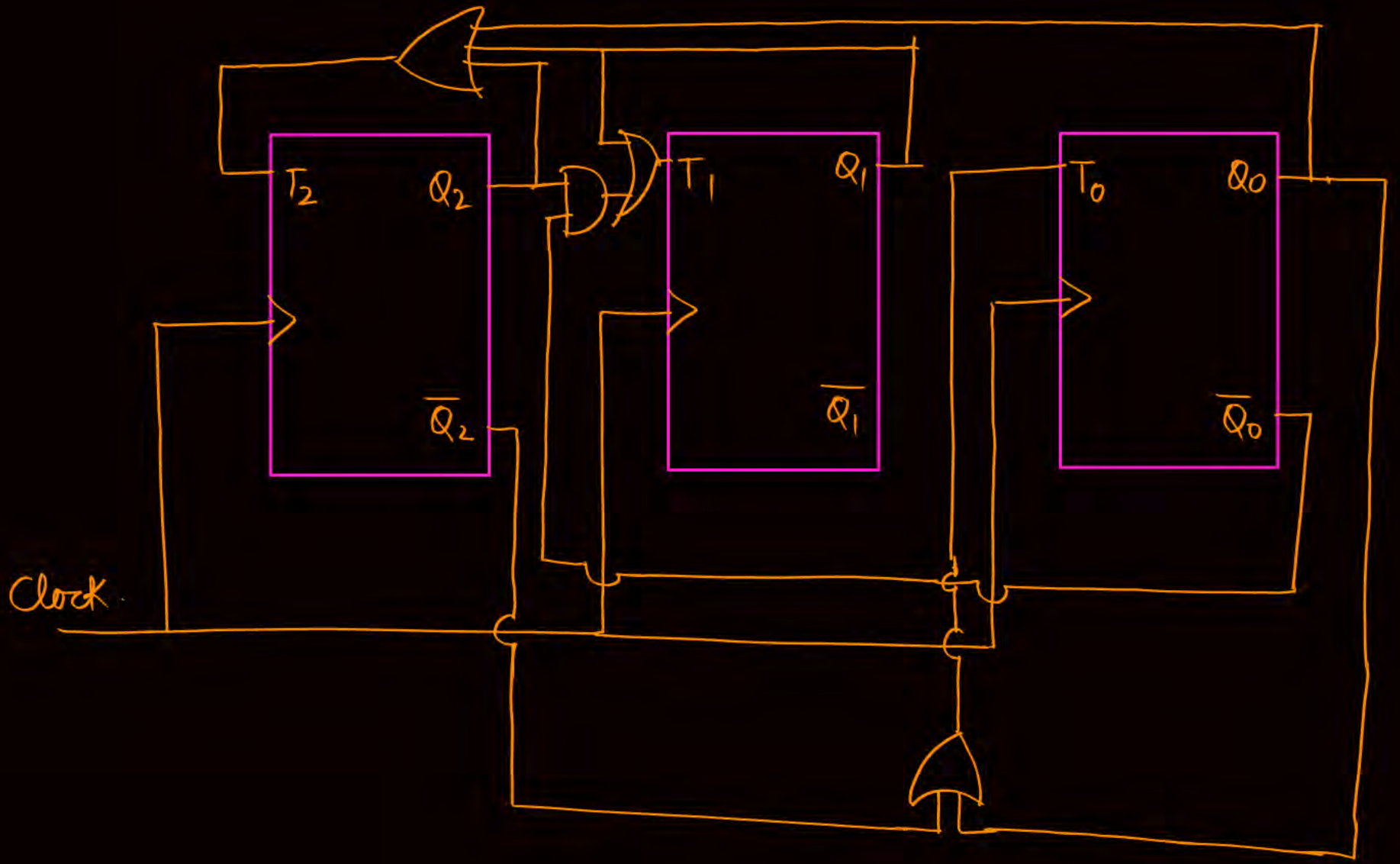
$$T_0 = \bar{Q}_2(n) + Q_0(n)$$



$$T_2 = Q_2(n) + Q_0(n) + Q_1(n)$$

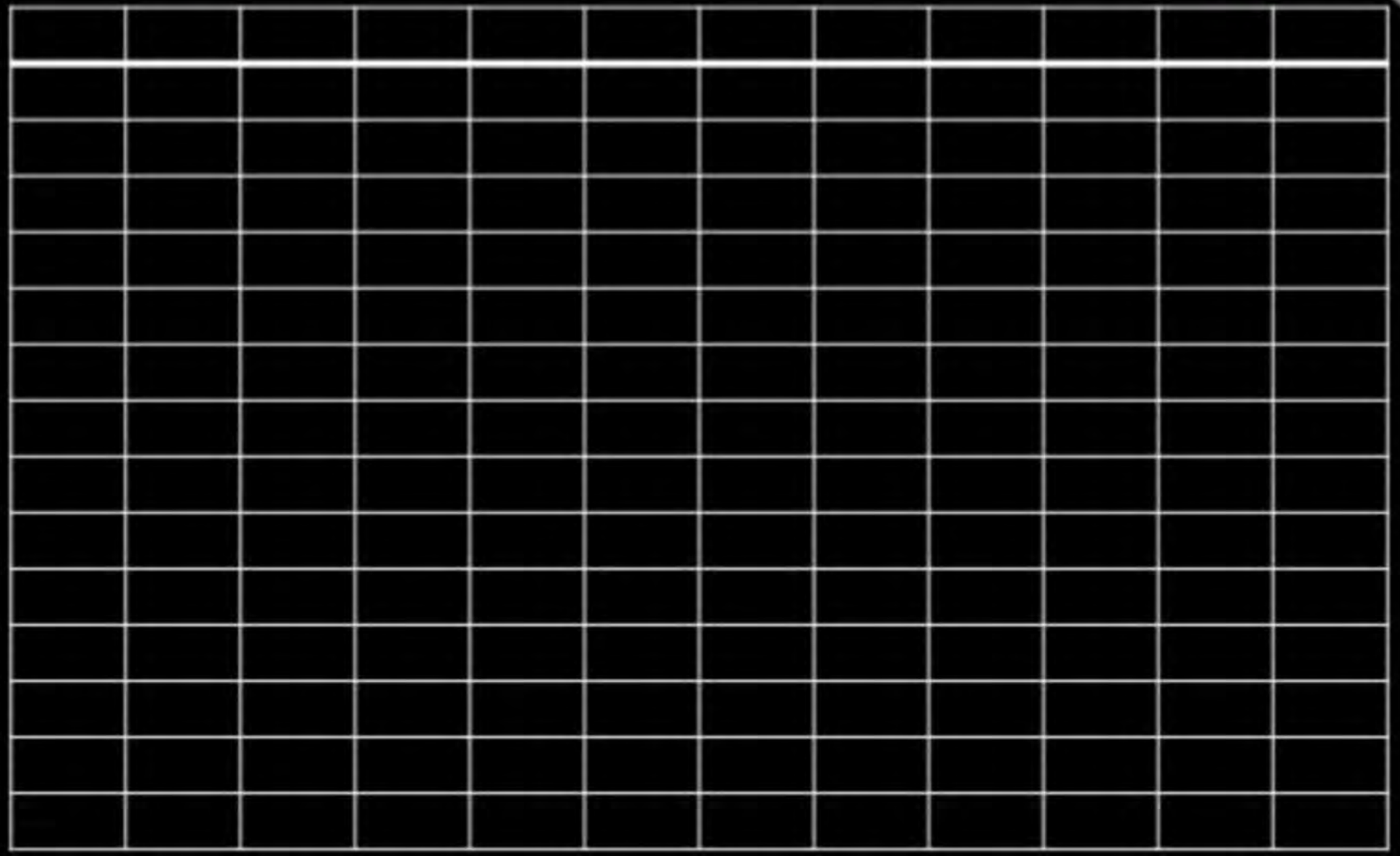


$$T_1 = Q_2\bar{Q}_0 + Q_1$$


T_2
Q_2
$\overline{Q_2}$
T_1
Q_1
$\overline{Q_1}$
T_0
Q_0
$\overline{Q_0}$
Clock

Design a Counter with Sequence: (1-2-4-8-12-14-15-0-3) using DFF.

H.W.



Design a Counter with sequence → (0-3-1-2-2) using TFF → 2FF

→ get saturated at '2' [using TFF]

Q_1	Q_0	Q_1^+	Q_0^+	T_1	T_0			
0	0	1	1	1	1			
0	1	1	0	1	1			
1	0	1	0	0	0			
1	1	0	1	1	0			

$T_1[Q_1(n), Q_0(n)]$

$= \Sigma(0,1,3)$

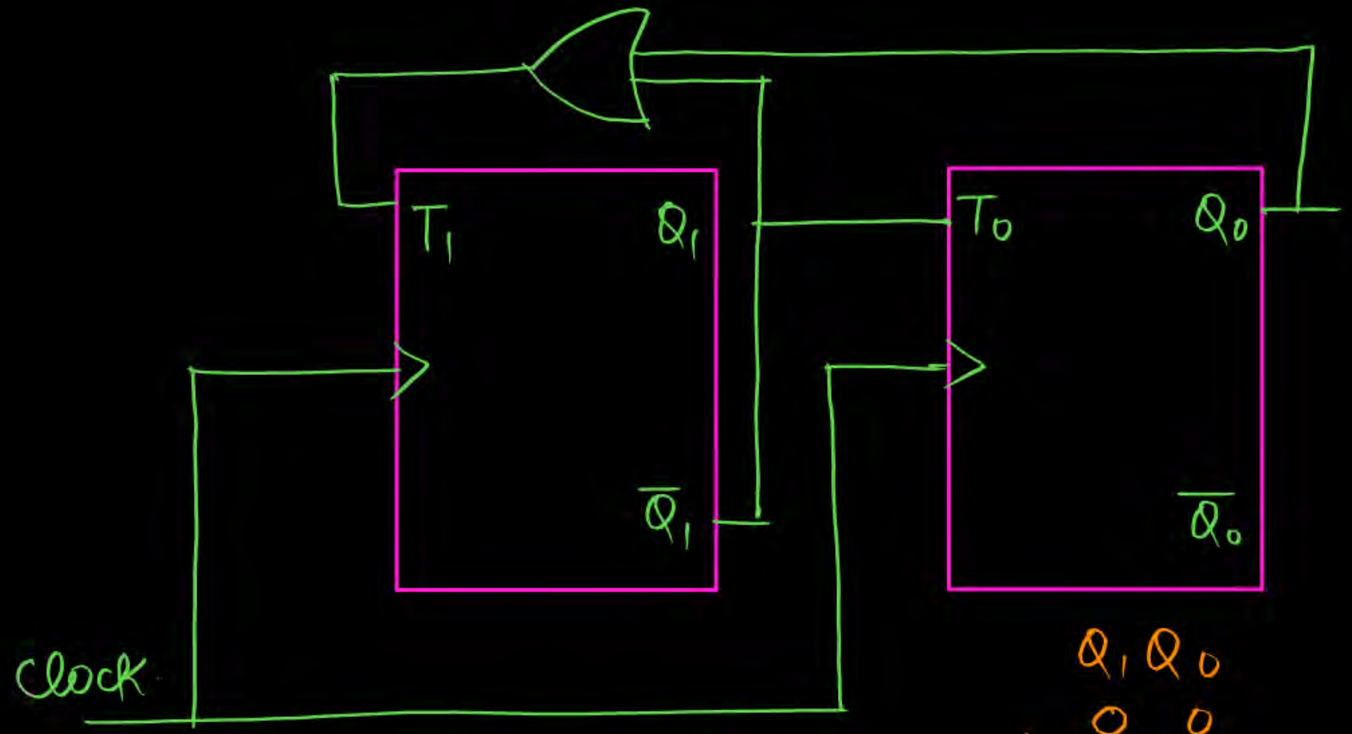
$= \pi(2)$

$= \overline{Q}_1(n) + Q_0(n)$

$T_0[Q_1(n), Q_0(n)] = \Sigma(0,1)$

$= \overline{Q}_1(n) \cdot \overline{Q}_0(n) + \overline{Q}_1(n) \cdot Q_0(n)$

$= \overline{Q}_1(n)$



$Q_1 Q_0 = (00)_2$

$Q_1(n+1) = T_1 \oplus Q_1(n)$

$= [\overline{Q}_1(n) + Q_0(n)] \oplus Q_1(n)$

$= \overline{Q_1(n)\cdot\overline{Q}_0(n)} \oplus Q_1(n)$

$= \overline{Q_1(n)\overline{Q}_0(n)} \oplus Q_1(n)$

$Q_1(n+1) = \overline{Q_1(n)[Q_0(n)]} \oplus Q_1(n)$

$Q_0(n+1) = T_0 \oplus Q_0(n) = \overline{Q}_1(n) \oplus Q_0(n)$

$= Q_1(n) \odot Q_0(n)$

Q_1	Q_0
0	0
1	1
0	1
1	0
1	0
1	0

Question

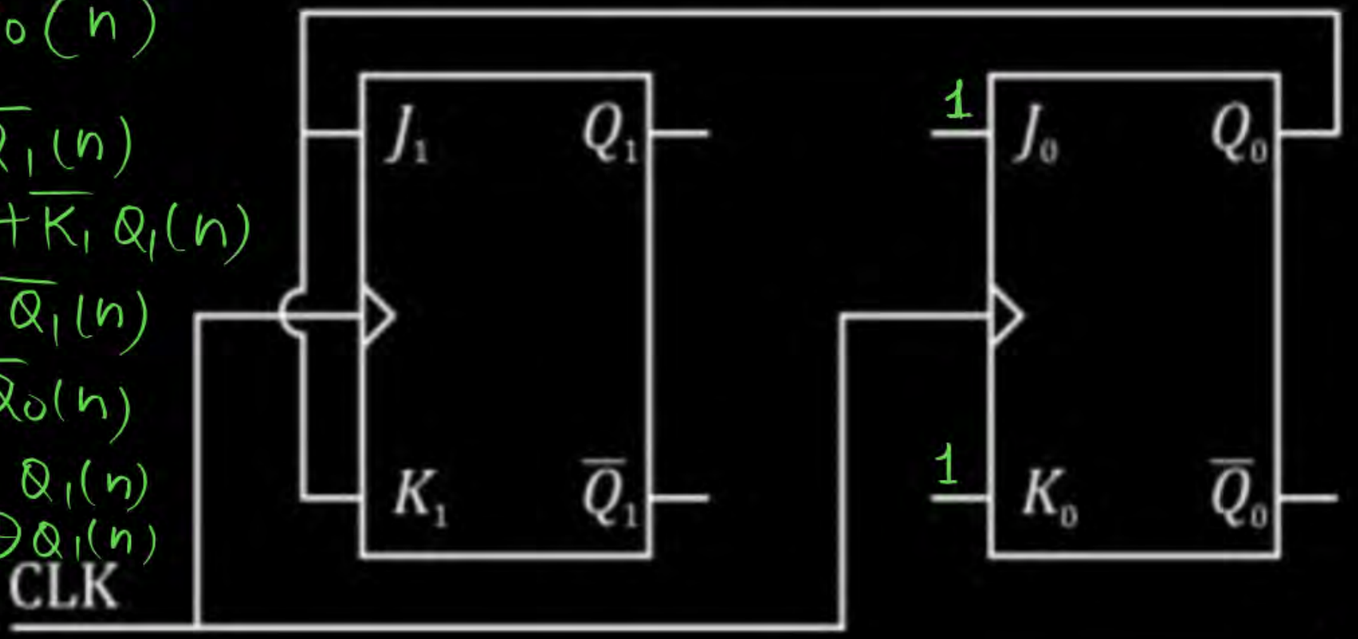
A sequential circuit is as given below :

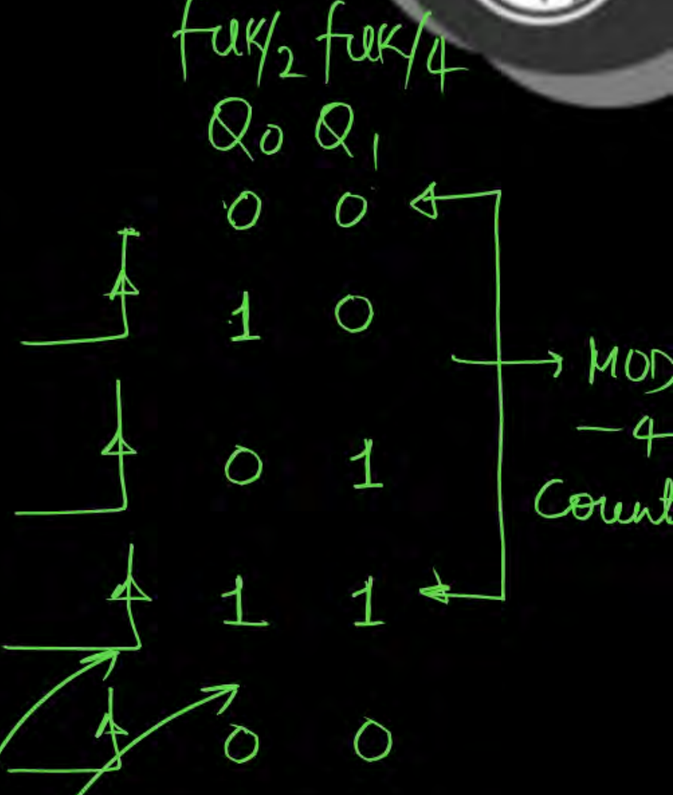
$Q_0(n+1) = \overline{Q_0}(n)$

$Q_1(n+1) = J_1\overline{Q_1}(n) + \overline{K_1}Q_1(n)$

$= Q_0(n)\overline{Q_1}(n) + \overline{Q_0}(n)Q_1(n)$

$= Q_0(n) \oplus Q_1(n)$





Initial state of the synchronous counter is given as $Q_0Q_1 = (00)_2$ what will be states of counter after applying clock pulses.

(a) 00 – 10 – 01 – 11 – 00

(b) 00 – 10 – 11 – 01 – 00

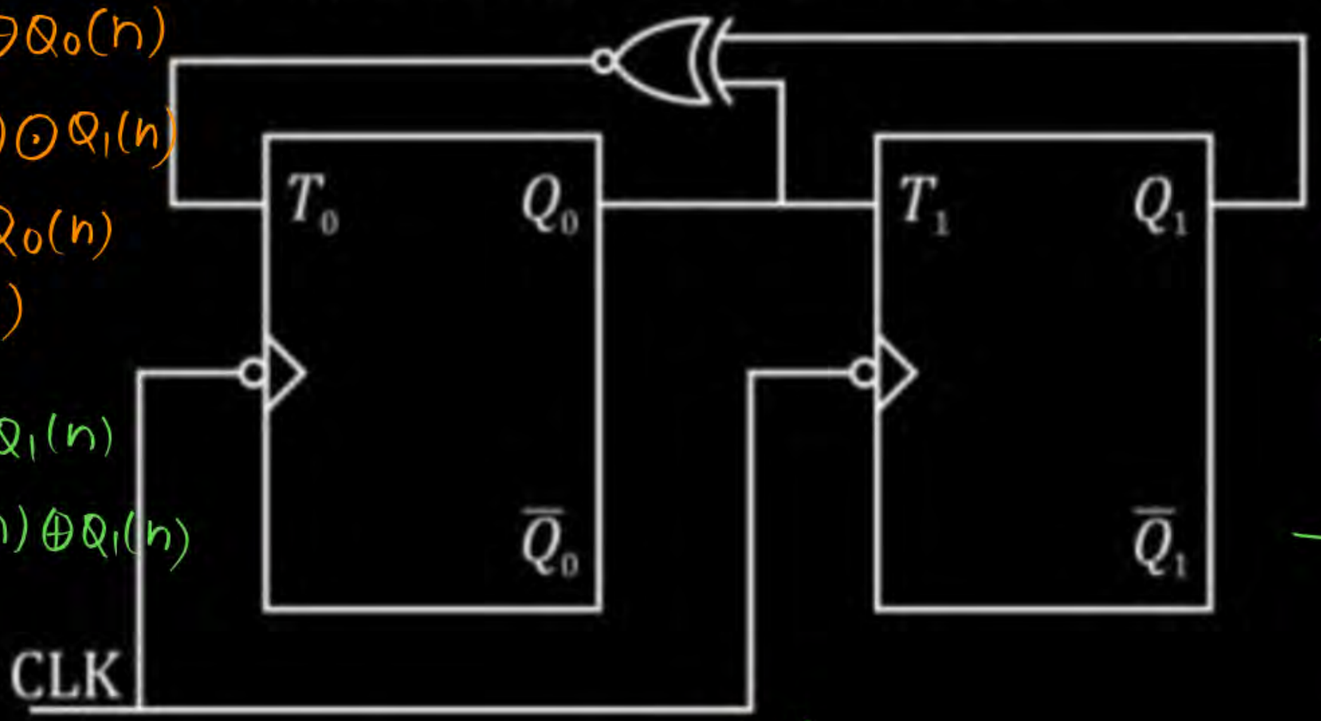
(c) 00 – 11 – 10 – 01 – 00

(d) 00 – 01 – 10 – 11 – 00

Question

The initial state of the given counter is $Q_1Q_0 = (00)_2$. What will be states of counter after applying clock pulses :

MSB LSB



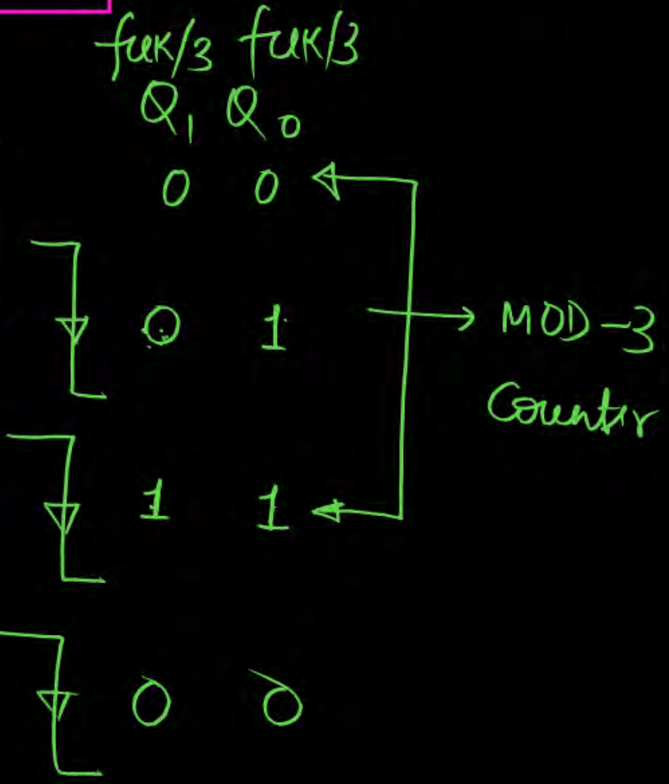
$Q_0(n+1) = T_0 \oplus Q_0(n)$

$= Q_0(n) \odot Q_1(n) \oplus Q_0(n)$

$Q_0(n+1) = \overline{Q_1}(n)$

$Q_1(n+1) = T_1 \oplus Q_1(n)$

$= Q_0(n) \oplus Q_1(n)$



(a) 00 – 10 – 11 – 01 – 00

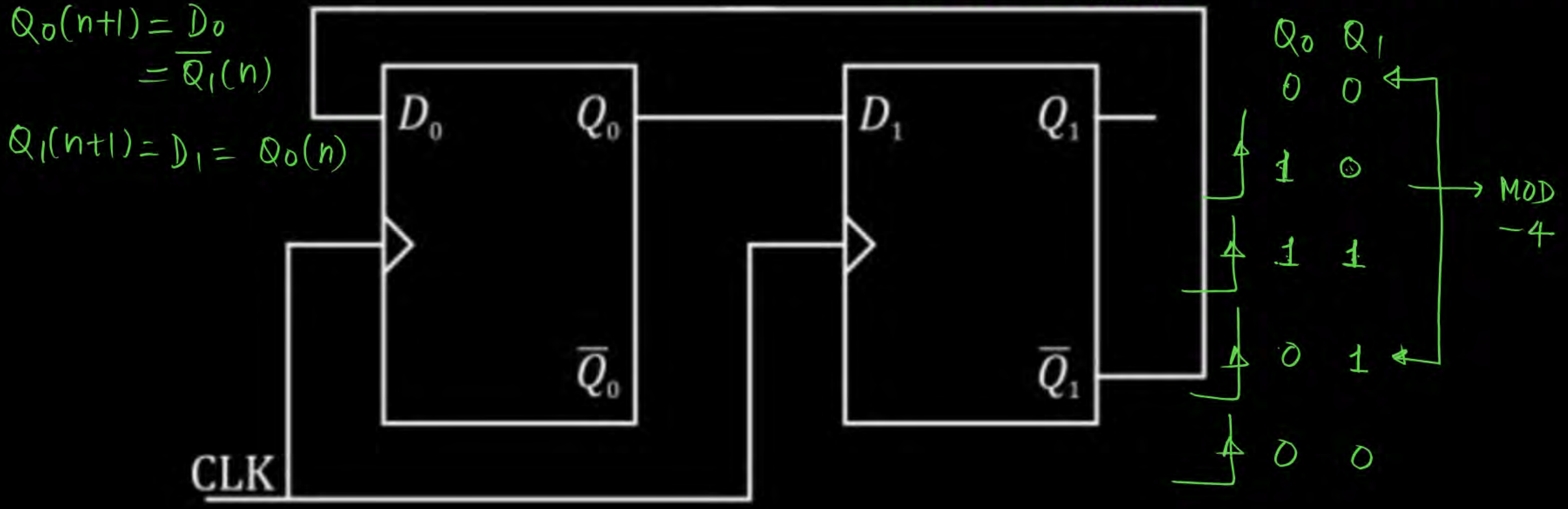
(b) 00 – 01 – 11 – 00

(c) 00 – 11 – 01 – 10 – 00

(d) 00 – 01 – 10 – 00

Question

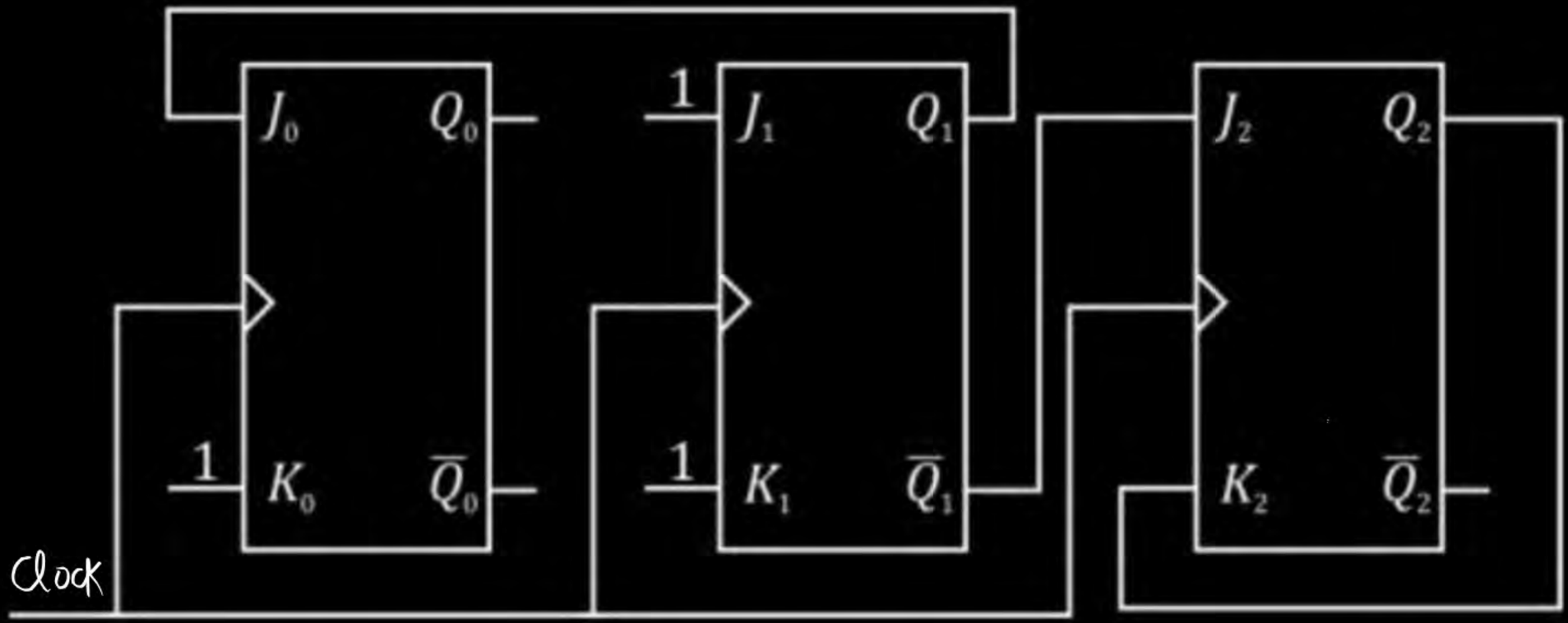
A sequential circuit is as given below :



Initial state of the counter is $Q_0Q_1 = (00)_2$. What will be the state of counter after application of 222 clock pulses $(3)_{10}$.

Question

A sequential circuit is as given below : H.W.



If initial state of counter is $Q_2Q_1Q_0 = (001)_2$, then after 8th clock pulse, counter will be in state ($)_{10}$.

Question

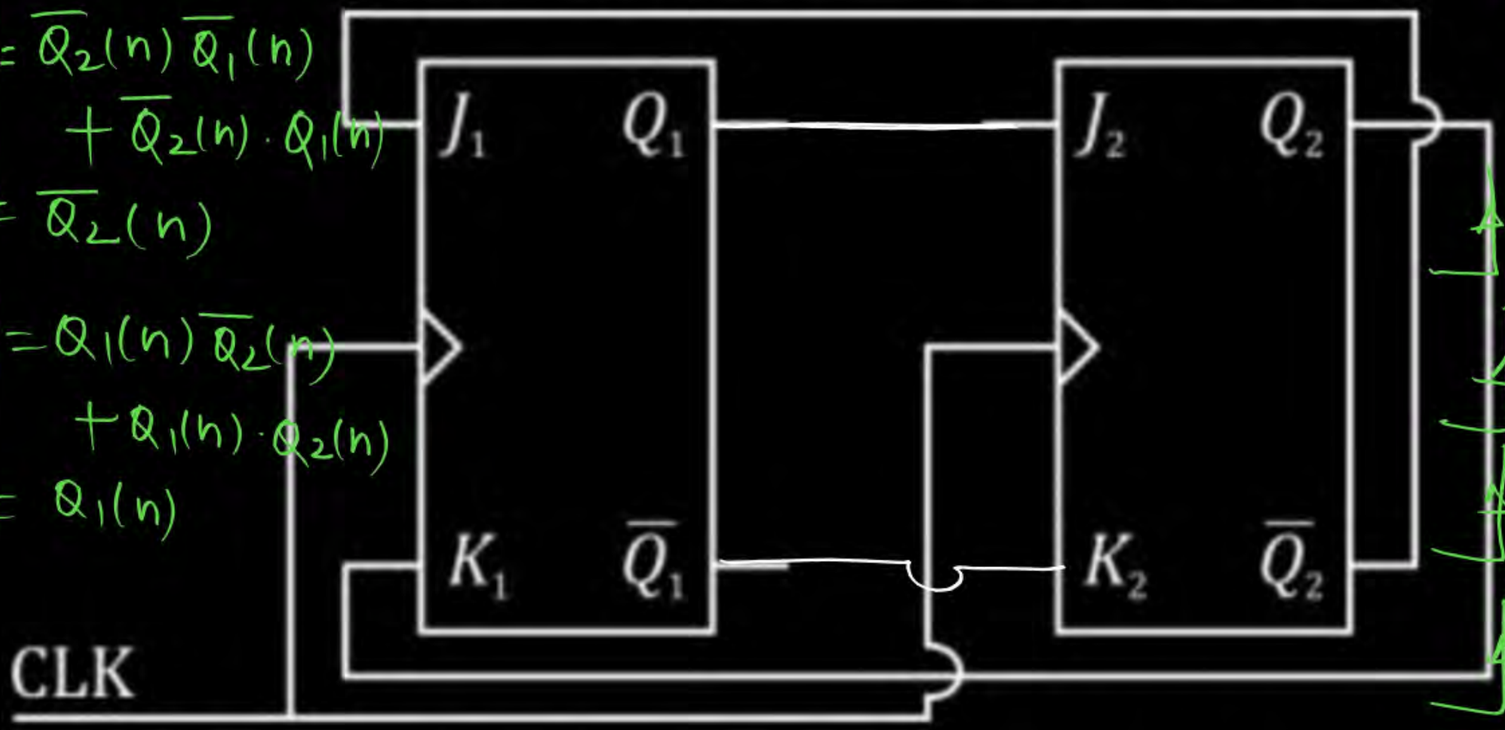
The output of the two FFs, Q_1 and Q_2 are initially at '0'. The sequence generated at Q_1 after application of clock pulses :

$Q_1(n+1) = \overline{Q_2}(n)\,\overline{Q_1}(n) + \overline{Q_2}(n) \cdot Q_1(n)$

$Q_1(n+1) = \overline{Q_2}(n)$

$Q_2(n+1) = Q_1(n)\,\overline{Q_2}(n) + Q_1(n) \cdot Q_2(n)$

$Q_2(n+1) = Q_1(n)$



Q_1	Q_2
0	0
1	0
1	1
1	0
0	1
0	0

Q_2	Q_1
0	0
0	1
1	1
1	0
0	0

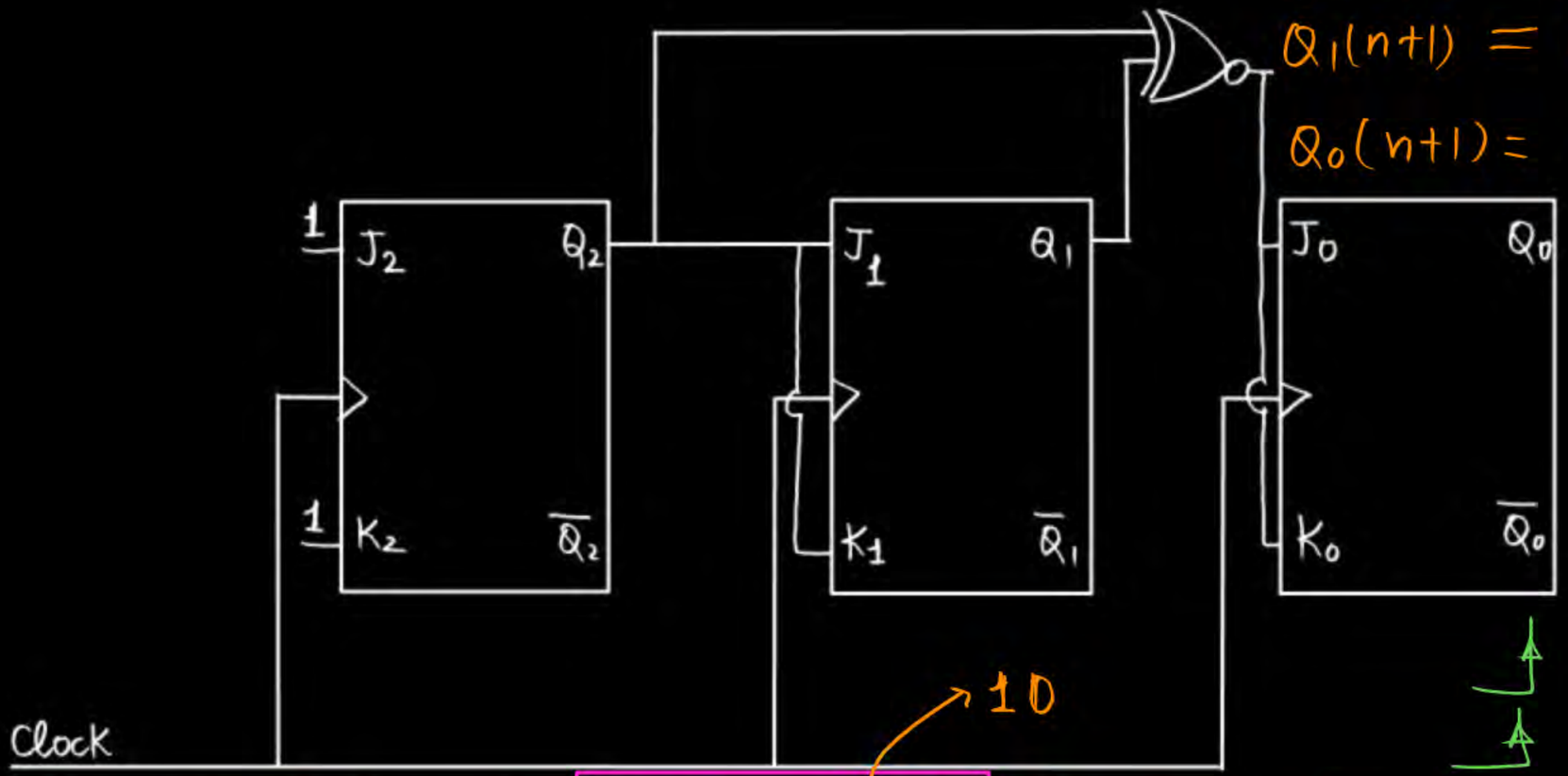
(a) 01110.....

(b) 01010.....

(c) 00110.....

(d) 01100.....

#Q. A sequential ckt is as given below:



$Q_2(n+1) = \overline{Q_2}(n)$

$Q_1(n+1) = Q_2(n)\overline{Q_1}(n) + \overline{Q_2}(n) \cdot Q_1(n)$

$Q_1(n+1) = Q_2(n) \oplus Q_1(n)$

$Q_0(n+1) = A\overline{Q}_0(n) + \overline{A}Q_0(n) = A \oplus Q_0(n)$

$= Q_2 \odot Q_1(n) \oplus Q_0(n)$

$= \overline{Q_2(n) \oplus Q_1(n) \oplus Q_0(n)}$

Q_2	Q_1	Q_0
0	0	0
1	0	1 ✓
0	1	1

10

The ckt starts with $Q_2Q_1Q_0 = (000)_2$ then next three states of the counter will be:

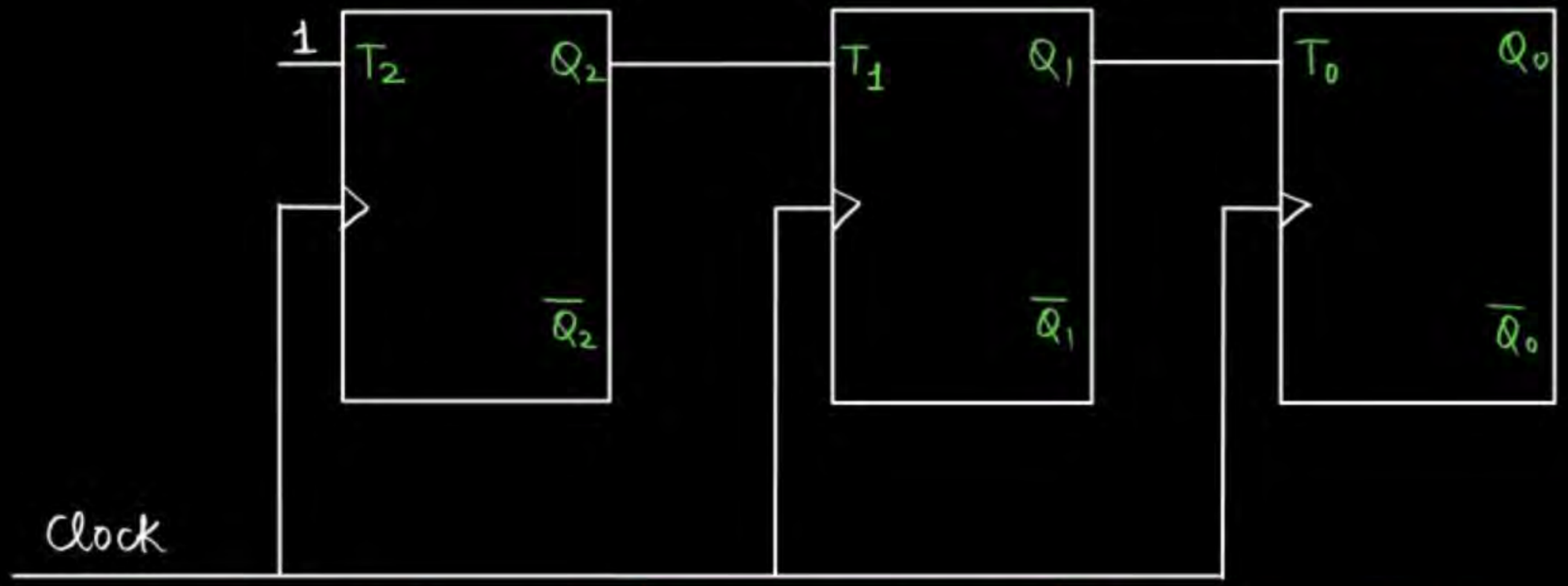
a. 101, 111, 011 b. 101, 010, 111 c. 101, 011, 111 d. 111, 011, 101

111

#Q. A sequential ckt is as given below:

$Q_2(n+1) = \overline{Q}_2(n)$ $\quad Q_1(n+1) = T_1 \oplus Q_1(n) = Q_2(n) \oplus Q_1(n)$

$Q_0(n+1) = Q_1(n) \oplus Q_0(n)$



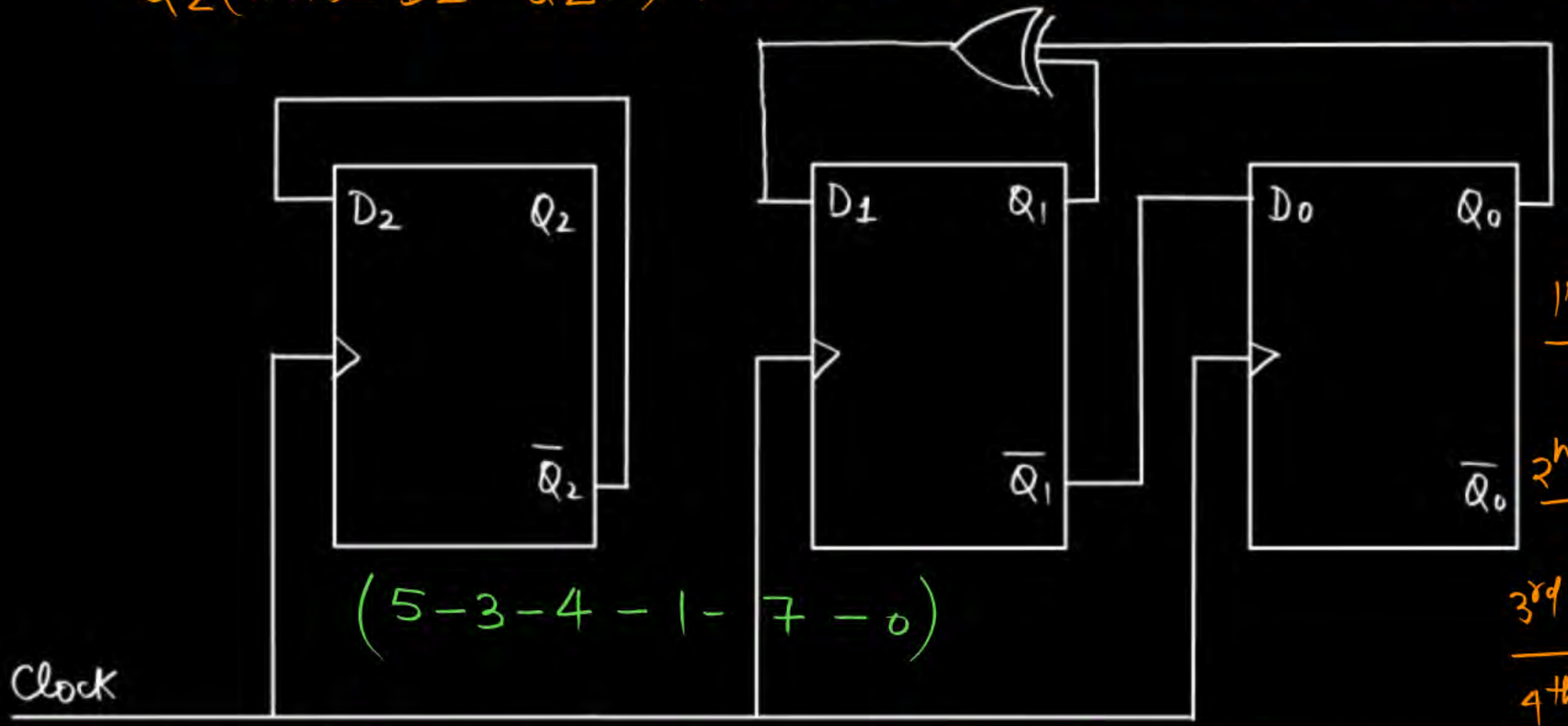
Q_2	Q_1	Q_0
1	0	0
0	1	0
1	1	1
0	0	0

The ckt starts with $Q_2Q_1Q_0 = (100)_2$ then state of the counter after 3rd clock will be

a. $(010)_2$ b. $(111)_2$ c. $(100)_2$ d. $(000)_2$

#Q. A sequential CKt is as given below:

$Q_2(n+1) = D_2 = \overline{Q_2}(n)$, $Q_1(n+1) = D_1 = Q_1(n) \oplus Q_0(n)$, $Q_0(n+1) = D_0 = \overline{Q_1}(n)$



ff's/2 ff's/3 ff's/3

$Q_2\ Q_1\ Q_0$

1 0 1

1st → 0 1 1

2nd → 1 0 0

3rd → 0 0 1

4th → 1 1 1

5th → 0 0 0

6th → 1 0 1

→ MOD-6

a. If starting state of the counter is $Q_2Q_1Q_0 = (101)_2$ then what is the state of the counter after 4th clock edge $\underline{(7)_{10}}$.

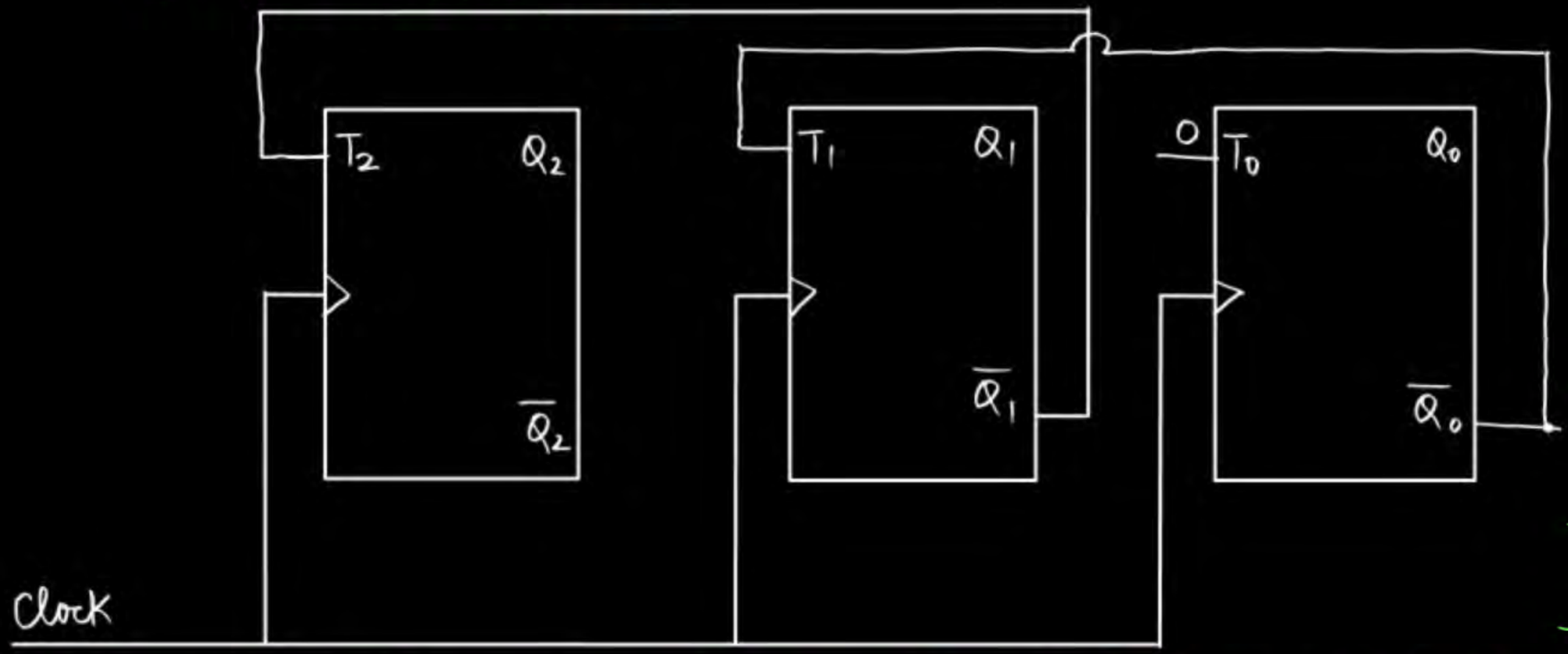
b. MOD no. of the counter $\underline{6}$.

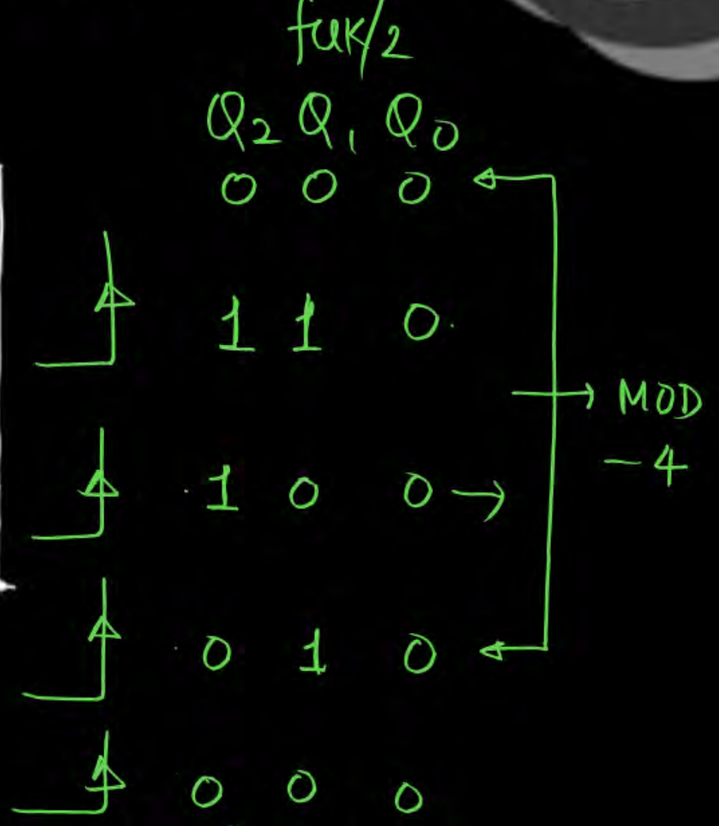
#Q. A sequential ckt is as given below:

$Q_2(n+1) = T_2 \oplus Q_2(n) = Q_1(n) \odot Q_2(n)$

$Q_1(n+1) = Q_0(n) \odot Q_1(n)$

$Q_0(n+1) = 0 \oplus Q_0(n) = Q_0(n)$



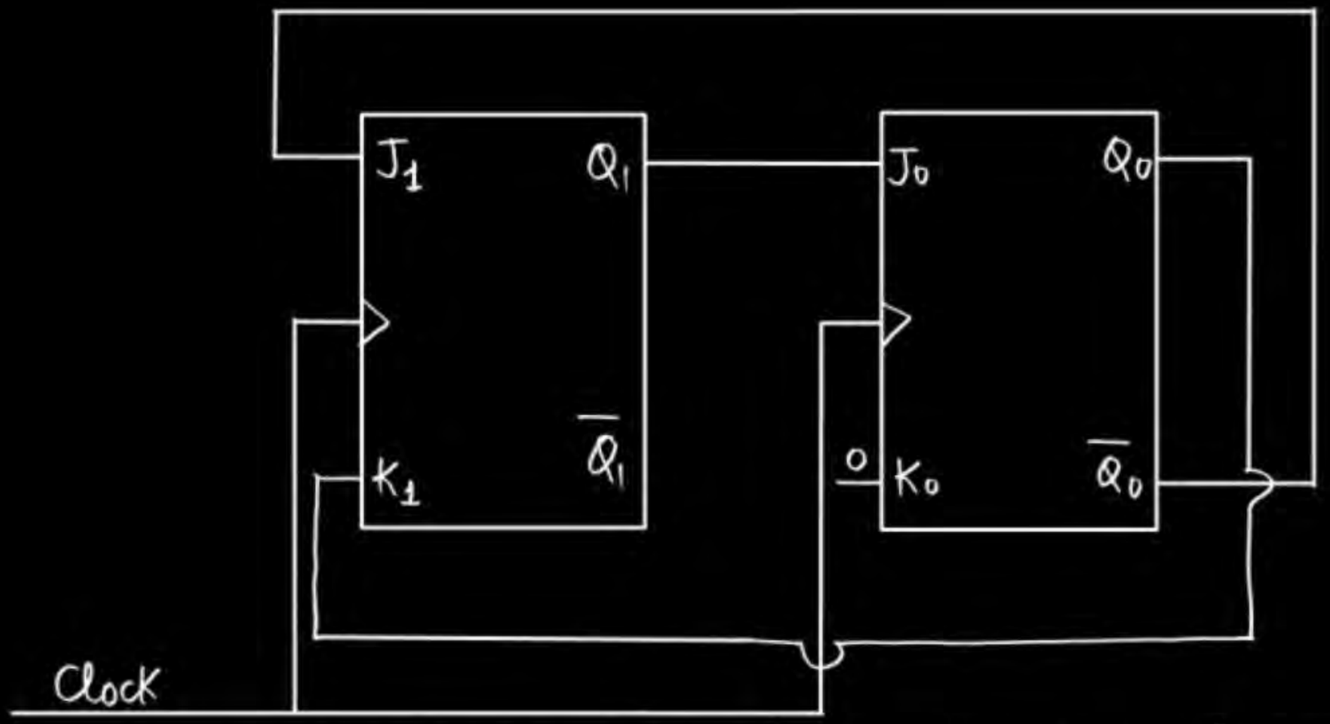


a. If the input clock frequency is 10 MHz, then o/p frequency at o/p Q_1 will be 5MHz MHz.

b. If counter starts with $(110)_2$ then summation of next three states will be $(6)_{10}$.

$Q_2 Q_1 Q_0 =$

#Q. A sequential CKt is as given below:



H.W

Given Counter started with $Q_1Q_0 = (00)_2$ then next four state of the counter will be:

a. 10, 11, 01, 00 b. 10, 11, 00, 10 C. 10, 11, 01, 01 d. 01, 11, 10, 10

- $(0-1-2-1-3-1-4) \longrightarrow$ MOD no. 7

Q_4	Q_3	Q_2	Q_1	Q_0	
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	2
0	1	0	0	1	9
0	0	0	1	1	3
1	0	0	0	1	17
0	0	1	0	0	4

- (0 1 1 2 2 2 3 3 3 3 4 4 4 4 4)

↓↓

6ff

000	000
000	001
001	001
000	010
001	010
010	010
000	011
001	011
010	011
011	011
000	100
001	100
010	100
011	100
100	100

- 0 - 1 - 2 - 0 - 1 - 3 - 0 - 1 - 4 - 2 - 4 - 3 - 4 - 1 - 3 - 1 - 5 - 7 - 9 - 3

7ff

1001 ⟶ 4 bit

for 5 different 1's ⟶ 3 bits



Topic : 2 Min Summary

Synch. Counter.

Thank you

GW

Soldiers !

